

GALL BLADDER & BILIARY TREE

Anatomy & physiology

- **GB:** 7–12 cm, 30–50 mL (distends to 300 mL). Parts: fundus, body, neck → **Hartmann's pouch** (infundibulum). **No submucosa/muscularis mucosa** (differs from GIT). Mucosal crypts = **crypts of Luschka**.
- **Cystic duct:** spiral mucosal folds = **valves of Heister**; sphincter of Lütken's.
- **Cystic artery:** 90% from right hepatic artery, in **Calot's triangle** (cystic duct, CHD, liver edge). **Moynihan's hump** = tortuous hepatic artery in front of cystic duct (danger in surgery).
- **CBD:** 7–11 cm; 4 parts (supraduodenal, retroduodenal, infraduodenal/pancreatic, intraduodenal); opens at **ampulla of Vater** via **sphincter of Oddi**.
- **Sphincter of Oddi:** regulates bile/pancreatic flow, prevents reflux, diverts bile to GB. Relaxes when **CCK** rises.
- Nerve: **vagus** = **stimulant**, sympathetic (celiac) = inhibitory.
- **Bile:** 500–1000 mL/day; 97% water + bile salts + pigment. CCK contracts GB + relaxes Oddi; **somatostatin inhibits GB contraction (↑gallstone risk)**.
- **Enterohepatic circulation:** primary bile salts (cholate, deoxycholate) → 80% reabsorbed in terminal ileum; gut bacteria → secondary bile acids.

Diagnostic studies

- **Bloods:** LFT (↑bilirubin + alk phos in cholangitis/cholestasis); **normal in biliary colic**.
- **Plain X-ray:** 10% radio-opaque stones, porcelain GB, limey bile.
- **Ultrasound = standard** for gallstones + jaundice.
- **HIDA scan:** acute cholecystitis, ampullary obstruction, post-op leaks.
- **MRCP = gold standard for biliary tree anatomy + CBD stones**.
- **ERCP:** diagnostic + therapeutic (papillotomy, stone removal, stenting). **PTC** if ERCP unavailable. EUS for periampullary/depth.
- **CT:** inferior to US for stones; used for malignancy.

Congenital anomalies

- GB anatomy classic in only 1/3. Absent GB (rare), **Phrygian cap** (2–6%), duplicated GB, floating GB (torsion risk), ducts of Luschka.
- Cystic/hepatic artery anomalies in **50%**.
- **Extrahepatic biliary atresia:** 1/4000; BD destroyed by inflammation → cirrhosis; jaundice from birth, pale stool, dark urine; **untreated dies before age 3**. Type 1 (CBD) → Roux-en-Y; Types 2,3 → **Kasai procedure**.
- **Caroli's disease:** congenital intrahepatic duct dilatation → stones, cholangitis.
- **Choledochal cyst:** congenital dilatation of biliary tree; **female 6:1**, eastern countries; >90% anomalous pancreaticobiliary junction; **premalignant** → excision + Roux-en-Y hepaticojejunostomy.

Gallstones (cholelithiasis)

- Commonest GIT surgery in west; rare Africa/India. **F:M 3:1** (equal in old age). **Saint's triad** = gallstones + hiatus hernia + colonic diverticulosis.
- Asymptomatic → 3%/yr become symptomatic; complicated 3–5%/yr.
- **Risk (5 F's):** Fat, Female, Forty, Fertile, Family. Also: obesity, pregnancy, ileal disease/resection, haemolytic disease (sickle, thalassemia, spherocytosis).

- **Mechanisms:** metabolic (cholesterol supersaturation = lithogenic bile), infection (E. coli, pigment stones), bile stasis.
- **Types:** **Cholesterol** (west, 80%, solitary/large, mostly radiolucent), **pigment** (<20% cholesterol — **black** = haemolysis/cirrhosis, in GB; **brown** = infection/parasites, in ducts, Asia).
- **Effects:** GB (silent, acute/chronic cholecystitis, mucocele, empyema, gangrene, perforation, carcinoma); ducts (obstructive jaundice, cholangitis, biliary pancreatitis); intestine (**gallstone ileus**).

Acute cholecystitis

- 90–95% from gallstones obstructing cystic duct → inflammation (lysolecithin, bile salts).
- **Clinical:** biliary colic NOT subsiding for days, RUQ/epigastric/back pain, fever, N/V; **Murphy's sign +ve**, guarding, ± mass.
- **Investigations:** ↑WBC (>20,000 = complication), mild ↑LFT; **US 95%** sonographic features; HIDA if atypical.
- **Treatment:** **conservative** (bed rest, IVF, analgesia, antibiotics, NGT) — **90% respond** → **interval cholecystectomy** at 6–10 wks. 10% no response or gangrene/perforation → **cholecystectomy** (laparoscopic/open); unfit → percutaneous cholecystostomy.

Chronic cholecystitis (biliary colic)

- Cystic duct obstruction → distension. Aschoff-Rokitansky sinuses. Pain 1–5h, after fatty/heavy meal, ± N/V, bloating/belching.
- Investigation: WBC/LFT normal; **US (stones, sludge)**.
- Treatment: avoid heavy/fatty meals; **cholecystectomy** if diabetic, atypical recurrent, or symptomatic. **Mucocele** (stone impacted, bile absorbed, mucus secreted → palpable non-tender GB) → cholecystectomy.

Other benign conditions

- **Cholecystoses:** cholesterosis (**strawberry GB** — yellow specks), cholesterol polyps, adenomyomatosis. Cholecystectomy if symptomatic/enlarging.
- **Empyema:** GB full of pus (sequel of cholecystitis/mucocele) → drainage + cholecystectomy.
- **Acalculous cholecystitis:** no stone; **after major surgery/trauma/burns/ICU**; missed → 20% mortality. → cholecystostomy.
- **Cholangitis:** ascending bacterial infection + biliary obstruction (E. coli, Klebsiella). **Charcot's triad** = pain + fever + jaundice (2/3); **Reynolds pentad** = + shock + confusion. Treat IVF + antibiotics; 15% need decompression (ERCP/PTC/surgery).
- **Choledocholithiasis:** 6–12% of gallstone patients. **Primary** (Asian, pigment, stasis) vs **secondary** (western, cholesterol, from GB). → obstructive jaundice, cholangitis, pancreatitis. **Courvoisier's law:** palpable GB + jaundice → NOT stone (think cancer). **MRCP gold standard**; treat **ERCP + papillotomy**.
- **Biliary pancreatitis:** stone at ampulla → ERCP + stone removal, then cholecystectomy.
- **Recurrent pyogenic cholangitis (cholangiohepatitis):** Orient; bacteria + parasites → brown pigment stones, strictures, abscess. MRCP/PTC; treat stones ± hepaticojejunostomy.
- **Sclerosing cholangitis:** inflammatory stricture intra/extrahepatic. Primary (unknown) or secondary; **2/3 associated with UC**; **cholangiocarcinoma risk 10–20%**. ERC = "beading"; treat stenting/transplant.
- **Parasites:** Ascaris (via ampulla), hydatid cyst, **Clonorchis sinensis** (BD carcinoma risk), **Fasciola hepatica** (water cress; eosinophilia; triclabendazole).

- **Typhoid GB** (Typhoid Mary = carrier), **torsion of GB** (elderly, long mesentery), **post-cholecystectomy syndrome** (15%; exclude retained CBD/stump stone, stricture).

Bile duct injury & stricture

- BD injury: mostly **iatrogenic** (cholecystectomy). Pain, fever, ↑LFT, bile leak/peritonitis, jaundice. Treat by drainage/stent → **Roux-en-Y hepaticojejunostomy**.
- BD stricture: 80% post-operative; unrecognized → recurrent cholangitis + secondary biliary cirrhosis.

Biliary tumours

- **Cholangiocarcinoma**: M=F, 50–70 yrs; risk = **PSC, UC, choledochal cyst, Clonorchis, hepatolithiasis, typhoid carrier, thorotrast**. Common at **hilar bifurcation = Klatskin tumour** (Bismuth-Corlette classification). **Painless jaundice + pruritus + weight loss**. Markers CA19-9, CEA. Resectable only 10–20%; median survival 18 mo.
- **Gallbladder carcinoma**: rare; **F:M 5:1**; **>90% have gallstones**; mostly adenocarcinoma. Risk: stones >3cm (10×), polyp >10mm, **porcelain GB**, choledochal cyst, anomalous junction. Spreads to liver early; 40% metastatic at diagnosis. Treat by cholecystectomy → extended cholecystectomy + hepatectomy depending on depth.

High-yield one-liners

- **Calot's triangle** (cystic duct, CHD, liver) → **cystic artery**. Hartmann's pouch, valves of Heister.
- **5 F's**; cholesterol stones (wet, radiolucent); **black** = haemolysis; **brown** = infection.
- **Murphy's sign** = acute cholecystitis; **US 95%**; **90% respond to conservative** → interval cholecystectomy.
- **Charcot's triad** (pain+fever+jaundice) = cholangitis; **Reynolds pentad** +shock+confusion.
- **Courvoisier's law**: palpable GB + jaundice ≠ stone (think cancer).
- **MRCP** = gold standard biliary tree/CBD stones; **ERCP** = therapeutic.
- **Choledochal cyst + PSC** = cholangiocarcinoma risk; **Klatskin** = hilar.
- **GB cancer**: **F 5:1**, **>90% gallstones**, **porcelain GB** risk, poor prognosis.
- **Biliary atresia**: Kasai procedure; untreated dies <3yr.